Meera Krishnamoorthy

Union City, NJ meerak@umich.edu 408 857 2695 meerakulous.github.io

Summary

I am an AI Research Engineer at Kira Learning, building the AI-powered systems that enable Kira to provide personalized learning for students. My research develops machine learning methods for real-world problems, with a focus on model evaluation, distribution shift, multiple instance learning, and survival analysis. My experience spans education and healthcare, including large medical image analysis and predicting time to spontaneous labor.

Technical Skills

Machine Learning: Supervised Learning, Multiple Instance Learning, LLM Evaluation, Personalized Learning, Model Evaluation, Survival Analysis, Distribution Shift, Convolutional Neural Networks, Transformers

Programming: Python (PyTorch, scikit-learn, NumPy), SQL

Education

University of Michigan *Aug. 2019 – June 2025*
Ph.D. in Computer Science

- GPA: 3.9/4.0
- Thesis Title: Towards Scalable and Stable Machine Learning in Clinical Contexts: Addressing Computational Efficiency and Dataset Shift
- Thesis Committee: Jenna Wiens (advisor), Emily Mower Provost, Elizabeth Bondi - Kelly, Michael Sjoding

California Institute of Technology *Sept. 2015 – June 2019*
B.S. in Electrical Engineering, Minor in Computer Science

- GPA: 3.7/4.0

Research and Work Experience

Kira Learning *New York City, NY*
AI Research Engineer *June 2025 – Present*

- Building the AI-powered systems that enable Kira to provide personalized learning for students: generating granular skills from academic standards, tagging educational content with those skills, estimating student mastery on those skills from performance on the tagged content, and generating personalized material to improve mastery of target skills.
- Designing studies to evaluate Kira's personalized learning system, including diagnostic assessments, instructional routing, and analyses of student learning outcomes.
- Designed, monitored, and analyzed an SAT Summer Program study that provided preliminary evidence to support the efficacy of Kira's personalized learning system
- Built evaluation tooling for skill tagging, enabling an approximately 35% reduction in tagging costs while checking for quality regressions.

University of Michigan *Ann Arbor, MI*
Graduate Student Research Assistant *Aug. 2019 – June 2025*

- Advised by Professor Jenna Wiens
- Developed novel machine learning models that have advanced several key subfields of machine learning (metagenomic sequence classification, multiple instance learning, model selection, and survival analysis) with applications in healthcare (large medical image classification, time to spontaneous labor prediction)
- Work has resulted in models that improved accuracy by up to 15%, enhanced memory efficiency by 14-18%, and increased speed by up to 10x compared to standard approaches.
- Work published in leading clinical journals (Nature Medicine, Nature Communications Biology), and machine learning conferences (KDD, CHIL, MLHC).

Duke Institute for Health Innovation (DIHI) *Durham, NC*
Research Analyst Intern *June 2023 – Aug. 2023*

- Developed multiple models to predict post-operative outcomes from invasive surgical procedures.

California Institute of Technology *Pasadena, CA*
Summer Undergraduate Research Fellow *June 2018 – Aug. 2018*

- Mentored by Professor Yisong Yue.
- Explored using domain knowledge from control theory to develop a safe and accurate machine learning-based controller.

Rocketship.vc *Los Altos, CA*
Software Engineering Intern *June 2017 – Aug. 2017*

- Created method to scrape and store information about startup investors and founders.
- Performed social network analysis to find trends among networks of successful venture personnel.

NASA Jet Propulsion Laboratory *Pasadena, CA*
Summer Undergraduate Research Fellow *June 2016 – Aug. 2016*

- Mentored by Dr. Glenn Orton.
- Performed mathematical modeling and spectral analysis to identify nature of astronomical impact on Jupiter.

Stanford University *Pasadena, CA*
Research Intern *2013 - 2015*

- Mentored by Professor Shripad Tuljapurkar.
- Created mathematical models to simulate habitat degradation.

Awards and Fellowships

KDD Student Travel Award	*2025*
National Physical Science Consortium Fellowship	*2019*
Arthur E. Lamel Memorial Summer Undergraduate Research Fellowship	*2018*
SanPietro Travel Prize Recipient	*2018*
National Merit Scholar	*2015*

Selected Publications

Meera Krishnamoorthy, Donna Tjandra, Divya M. Shanmugam, Amanda E. Kowalski, Jenna Wiens. "Survival Analysis with Limited Overlap and Censoring Distribution Shift," Conference Paper, Machine Learning for Healthcare (MLHC), Aug. 2026.

Meera Krishnamoorthy, Michael W. Sjoding, Jenna Wiens. "Cross-Validation for Longitudinal Datasets with Unstable Correlations," Conference Paper, ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), Aug. 2025.

Meera Krishnamoorthy, Jenna Wiens. "Multiple Instance Learning with Absolute Position Information," Conference Paper, AHLI Conference on Health, Inference, and Learning (CHIL), June 2024.

Meera Krishnamoorthy, Michael W. Sjoding, Jenna Wiens. "Off-label use of artificial intelligence models in healthcare," Comment, Nature Medicine, March 2024.

Meera Krishnamoorthy, Piyush Ranjan, John R. Erb-Downward, Robert P. Dickson, Jenna Wiens. "AMAISE: a machine learning approach to index-free sequence enrichment," Journal Paper, Communications Biology 5, Article Number 568, June 2022.

Selected Presentations

Meera Krishnamoorthy, Jenna Wiens, "Improving the Memory Efficiency and Speed of Clinical Decision Support Tools," Oral Presentation, Greenhills Advanced Research Program: Summer Seminar Schedule Symposium, June 2024.

Meera Krishnamoorthy, Piyush Ranjan, John Erb-Downward, Robert Dickson, Jenna Wiens, "AMAISE: a machine learning approach to index-free sequence enrichment," Oral Presentation, University of Michigan's DCMB Tools & Technology Seminar, Feb. 2023.

Meera Krishnamoorthy, Piyush Ranjan, John Erb-Downward, Robert Dickson, Jenna Wiens, "AMAISE: a machine learning approach to index-free sequence enrichment," Poster Presentation, 10th Annual Kahn Symposium, Nov. 2022.

🏆 Honorable Mention Award 🏆

Meera Krishnamoorthy, Piyush Ranjan, John Erb-Downward, Robert Dickson, Jenna Wiens, "Machine Learning for Host Depletion of Metagenomic Data in Clinical Diagnostics," Poster Presentation, Machine Learning in Computational Biology (MLCB) 2021, Nov. 2021.

Teaching Experience

Artificial Intelligence for All (AI4All) *Ann Arbor, MI*
Instructor *2020 - 2021*

- Developed and helped students with Natural Language Processing projects involving translation and pun generation (Summer 2020)
- Taught coursework on data visualization, feature processing, and k-means clustering (Summer 2021)

California Institute of Technology *Pasadena, CA*
Undergraduate Teaching Assistant *2018 - 2019*

- CS/CNS/EE 156a: Learning Systems (Fall 2018)
- CS/CNS/EE 155: Machine Learning and Data Mining (Winter 2019)
- CS/CNS/EE 156b: Learning Systems Project Course (Spring 2019)

Professional/Academic Experience

Reviewer *2021 - present*

- ML4H 2024, Research2Clinics NeurIPS Workshop 2021, MLHC 2022, ML4H 2022, MLHC 2023, MLHC 2024, MLHC 2025

ML4H Career Mentorship Subchair *2024 - 2025*

- Matched mentors and mentees for career development support and organized virtual and in-person roundtables featuring experts in Machine Learning for Healthcare from industry and academia.

Michigan Science Center Coordinator *2023 - 2025*

- Developed AI-related exhibit for the Michigan Science Center.

Ensemble of Computer Science Ladies (ECSEL+) Social Chair *2021 - 2023*

- Planned social activities for members of ECSEL+

Michigan AI Symposium Poster/Demo Session Co-chair *2020*

- Helped advertise and organize the Michigan AI Symposium, a day of research talks, demos, and posters that brings together AI enthusiasts from industry and academia.

Caltech Undergraduate Research Journal Co-editor in Chief *2015 - 2019*

- Oversee editing and publication process of journal.

Volunteer Work

Caltech Robogals *Pasadena, CA*
Member *2015 - 2019*

- Teach robotics workshops to 1st - 8th grade students.

Caltech Society of Women Engineers *Pasadena, CA*
Member *2015 - 2019*

- Mentor younger members about classes and internships.
- Volunteer in community outreach events.

RISE Program *Pasadena, CA*
Tutor *2015 - 2019*

- Tutor 8th - 12th grade students in various math and science courses.